

FINANCIAL MARKET ANALYSIS ENGINE

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Abstract

Financial markets generate massive volumes of high-frequency time-series data characterized by volatility clustering, regime shifts, and complex inter-asset dependencies. Traditional CPU-based analytical pipelines struggle to process such data in real time, particularly when performing large-scale feature engineering, correlation analysis, signal transformation, and machine learning simultaneously. This project presents the design and implementation of a **GPU-Accelerated Financial Market Regime and Risk Intelligence System** built using the RAPIDS ecosystem and CUDA-enabled parallel computing.

The system leverages GPU-accelerated data wrangling using cuDF and NVTabular for large-scale feature engineering, including rolling volatility, momentum indicators, liquidity proxies, and drawdown metrics. Frequency-domain analysis is performed using cuSignal to detect volatility bursts and structural regime transitions through Fast Fourier Transform (FFT) and spectral density estimation. Market structure is modeled as a dynamic correlation graph using cuGraph, enabling detection of sector clustering, systemic risk concentration, and correlation breakdowns.

Machine learning models, including cuML-based Random Forest, Logistic Regression, and distributed XGBoost with Dask, are trained to classify market regimes and predict short-term volatility spikes. Multi-GPU parallelism and RAPIDS Memory Manager are employed to optimize computational efficiency and scalability. Comprehensive benchmarking compares CPU-based implementations (pandas, scikit-learn, SciPy) with GPU-accelerated pipelines in terms of execution time, memory usage, and throughput.

Rather than attempting deterministic price prediction, the system focuses on probabilistic risk detection and structural instability monitoring. The resulting framework demonstrates how GPU-accelerated data science can enable real-time financial intelligence, scalable risk analytics, and adaptive market monitoring.